# IGCSE 0607 Practice Questions — E9.3

## Probability of combined events

**Total Questions:** 6 — **Total Marks:** 38 Marks

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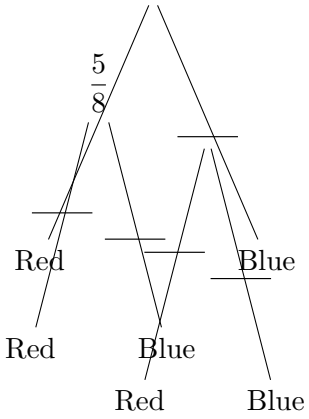
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**1** A bag contains 5 red counters and 3 blue counters.
A counter is taken at random from the bag, its colour is noted and it is replaced.
A second counter is then taken at random.

**(a)** Complete the tree diagram.



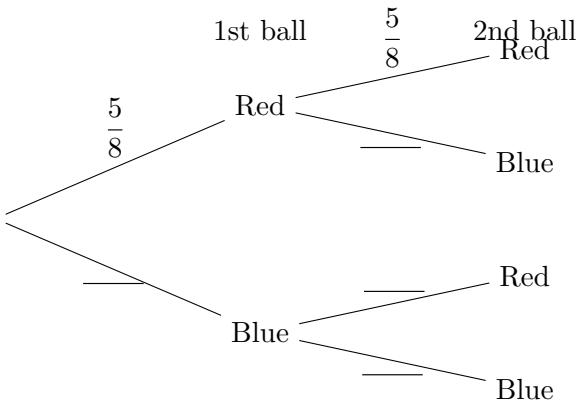

**[2]**

**(b)** Calculate the probability that both counters are red.

..... **[2]**

**2** A bag contains 8 balls.
5 of the balls are Red and 3 of the balls are Blue.
A ball is taken at random from the bag, its colour is noted, and it is replaced.
A second ball is then taken at random from the bag.

**(a)** Complete the tree diagram.




**[2]**

**(b)** Calculate the probability that both balls are Red.

..... **[1]**

**(c)** Calculate the probability of picking one Red ball and one Blue ball in any order.

..... **[2]**

**3** A box contains $n$ counters.
3 of the counters are Red and all the remaining counters are Blue.
Two counters are taken at random from the box, without replacement.
The probability that both counters are Red is $\frac{1}{15}$.

**(a)** Show that $n^2 - n - 90 = 0$.

**[3]**

**(b)** Find the value of $n$.

$n =$ ..... **[2]**

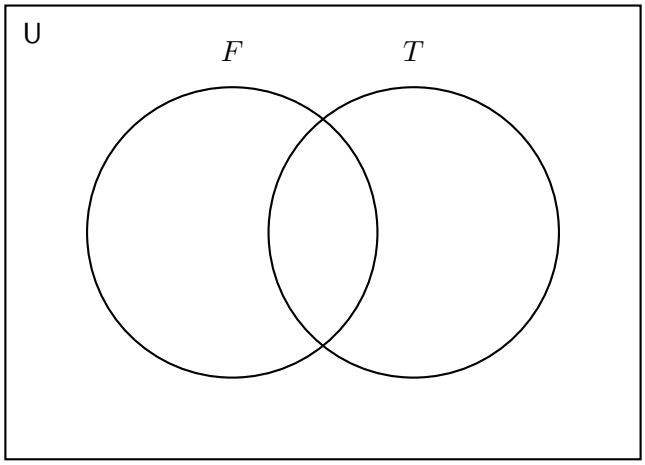
**(c)** Calculate the probability that exactly one of the two counters chosen is Red.

..... **[3]**

**4** There are 30 students in a class.
$\mathsf{U}$ = {students in the class}
$F$ = {students who play Football}
$T$ = {students who play Tennis

18 students play Football. 14 students play Tennis. 4 students do not play Football and do not play Tennis.

**(a)** Complete the Venn diagram to show the number of students in each region.




**[2]**

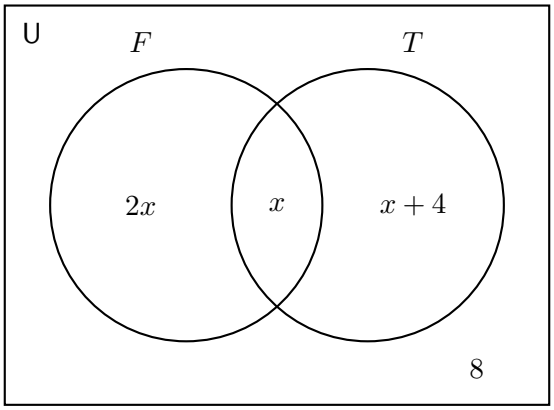
**(b)** A student is chosen at random from the class.
Find $P(F \cap T)$.

$P(F \cap T) =$ ..... **[1]**

**(c)** Two students are chosen at random from the class.
Calculate the probability that both students play Football but not Tennis.

..... **[3]**

**5** A group of 40 students are asked if they play Football ($F$) or Tennis ($T$).
The Venn diagram shows the number of students in each region in terms of $x$.




**(a)** Find the value of $x$.

$x =$ ..... **[2]**

**(b)** Find $P(F \cup T)$.

..... **[1]**

**(c)** Two students are chosen at random from the group of 40.
Calculate the probability that both students play Football only.

..... **[3]**

**6** A bag contains 4 red discs and some blue discs.
Let the total number of discs in the bag be $x$.
Two discs are taken at random from the bag, without replacement.
The probability that both discs are red is $\frac{1}{11}$.

**(a)** Show that $x^2 - x - 132 = 0$.

**[3]**

**(b)** Find the number of blue discs in the bag.

..... **[3]**

**(c)** Calculate the probability that one disc is red and one disc is blue.

..... **[3]**